

Chapter 6

Video Compression

IMP302m

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6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- Video compression is essential for the growth and success of low bandwidth transmission and storage of digital video.
- It is crucial for digital video communications, storage, processing, acquisition, and reproduction.
- Ensuring high-quality multimedia transmission over high-speed computer networks is a key challenge in designing quality of service (QoS) for digital transmission providers.

6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- Video compression reduces storage and transmission bandwidth needs, but it also impacts various system performance tradeoffs.
- When designing or selecting a video encoder, factors beyond compression efficiency must be considered, including bit rate versus distortion, algorithm complexity, transmission channel characteristics, algorithm symmetry, video source statistics, fixed versus variable rate coding, and standards compatibility.
- These considerations are crucial for making informed encoder design decisions.

6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- Coding efficiency = (compression ratio)⁻¹
= encoded bit rate/decoded bit rate
- The loss information or distortion measure is evaluated using mean square error (MSE), mean absolute error (MAE) or peak signal-to-reconstruction noise (PSNR)

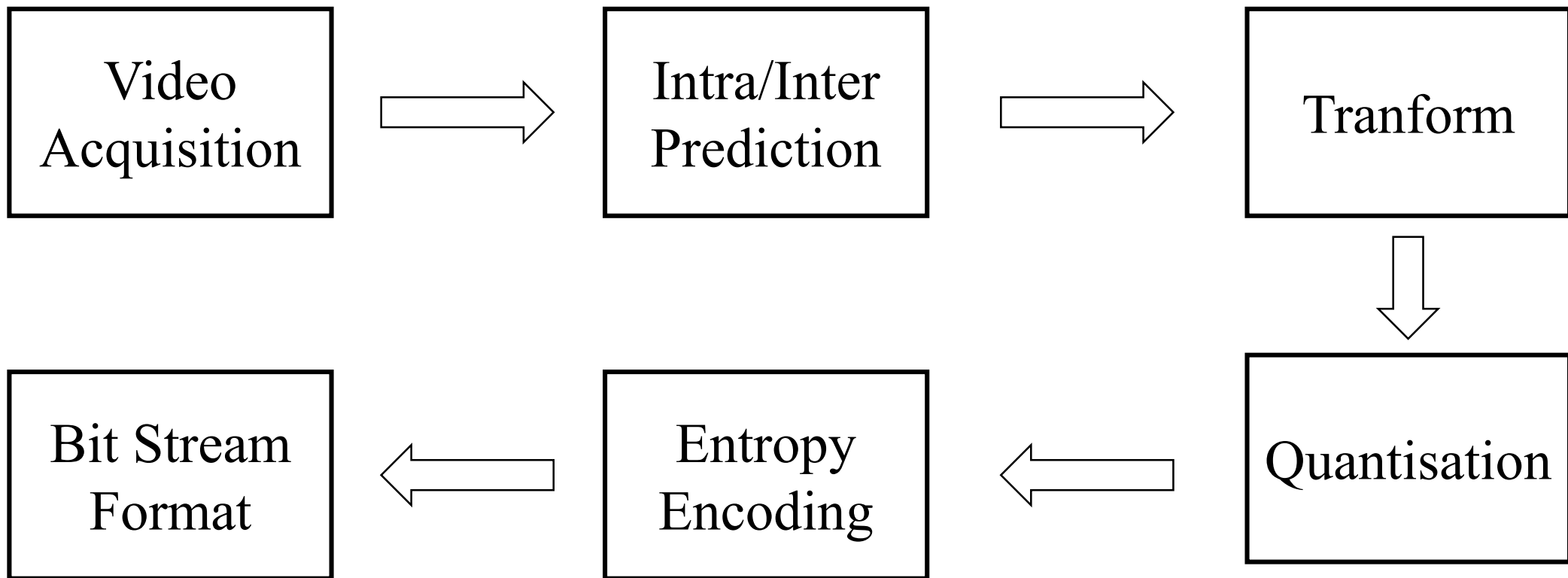
$$MSE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [I(i, j) - \hat{I}(i, j)]^2$$

$$MAE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N |I(i, j) - \hat{I}(i, j)|$$

$$PSNR = 20 \log_{10} \left(\frac{2^n}{MSE^{1/2}} \right),$$

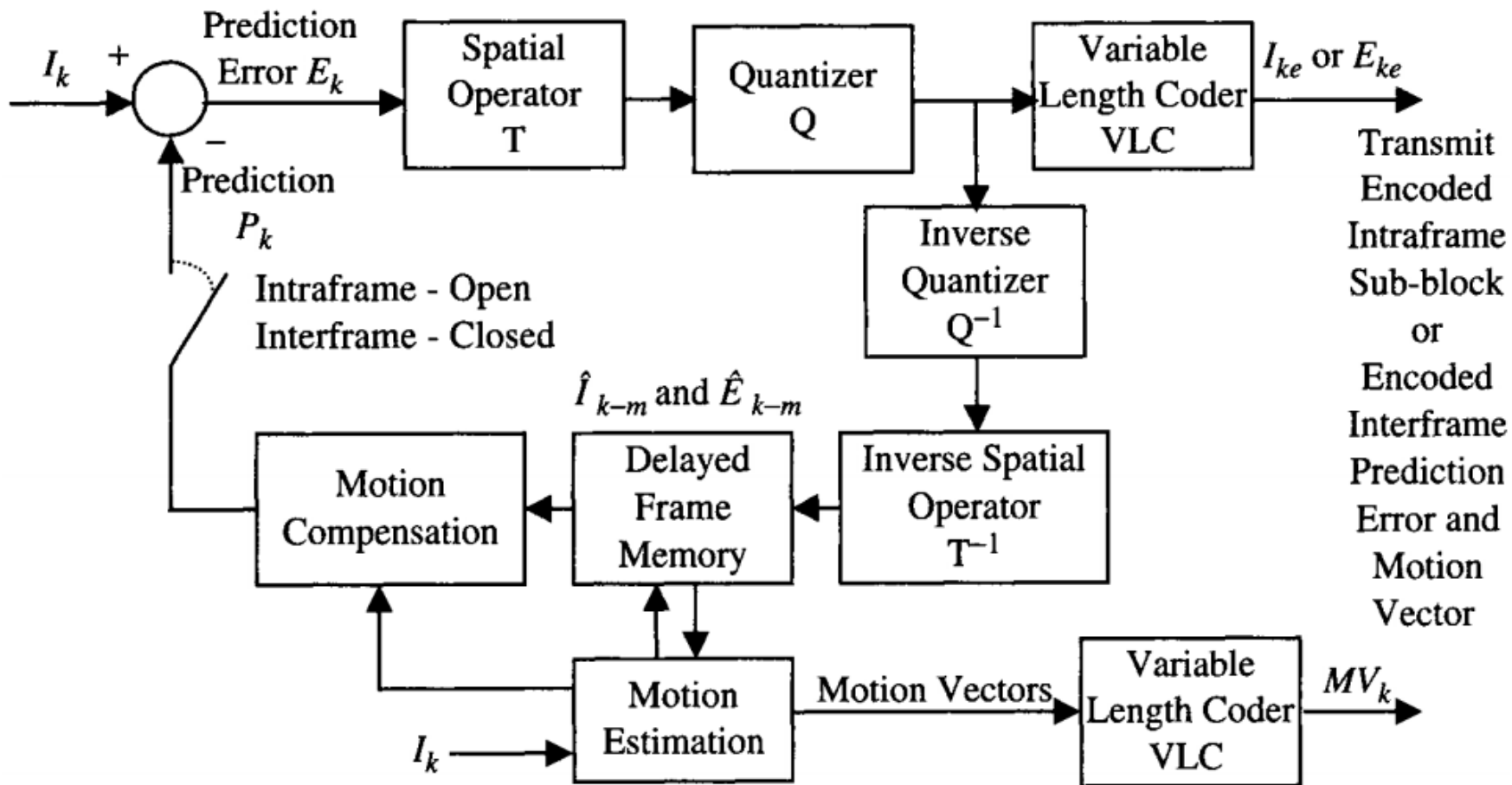
6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- Generalized video compression system



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6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

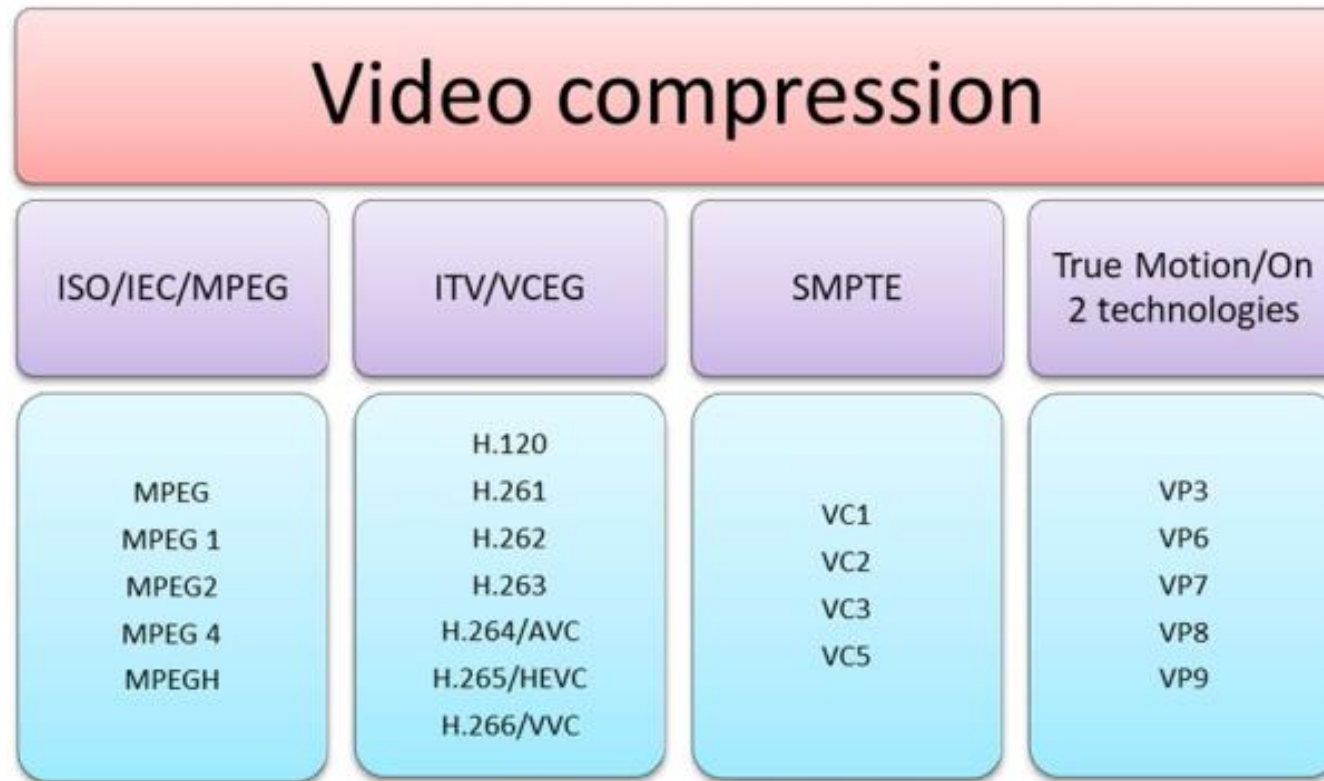
- Spatial operator: a unitary two dimensional linear transform, but in principle can be any unitary operator that can distribute most of the signal energy into a small number of coefficients, i.e., decorrelate the signal data.
- Quantizer - The spatial or transform operator is applied to the input in order to arrange the signal into a more suitable format for subsequent lossy and lossless coding operations.
- Inverse operators The inverse operators Q^{-1} and T^{-1} are applied to the encoded current frame I_{ke} or the current prediction error E_{ke} in order to reconstruct and store the encoded frame for the motion estimator and motion compensator to generate the next prediction frame

6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- Delayed frame memory: both current and previous frames must be available to the motion estimator and motion compensator to generate a prediction frame.
- Motion estimation: temporal encoding aspect of this system relies on the assumption that rigid body motion is responsible for the differences between two or more successive frames
- Motion compensator: makes use of the current frame motion estimates and the previously reconstructed frame to generate the current frame MV_k prediction

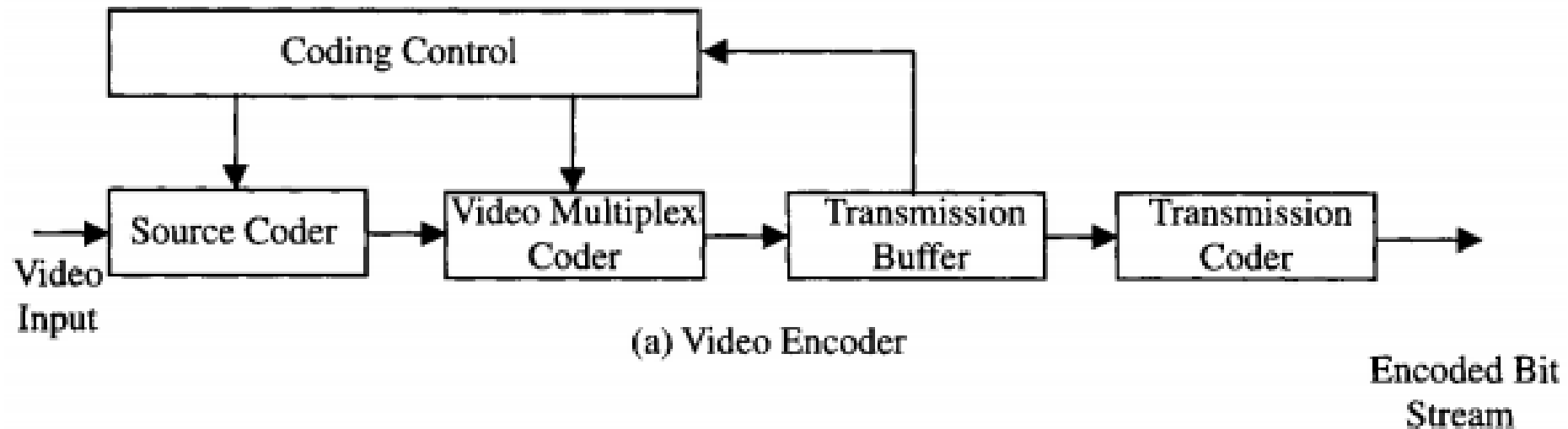
6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- Recapitulation of video compression techniques



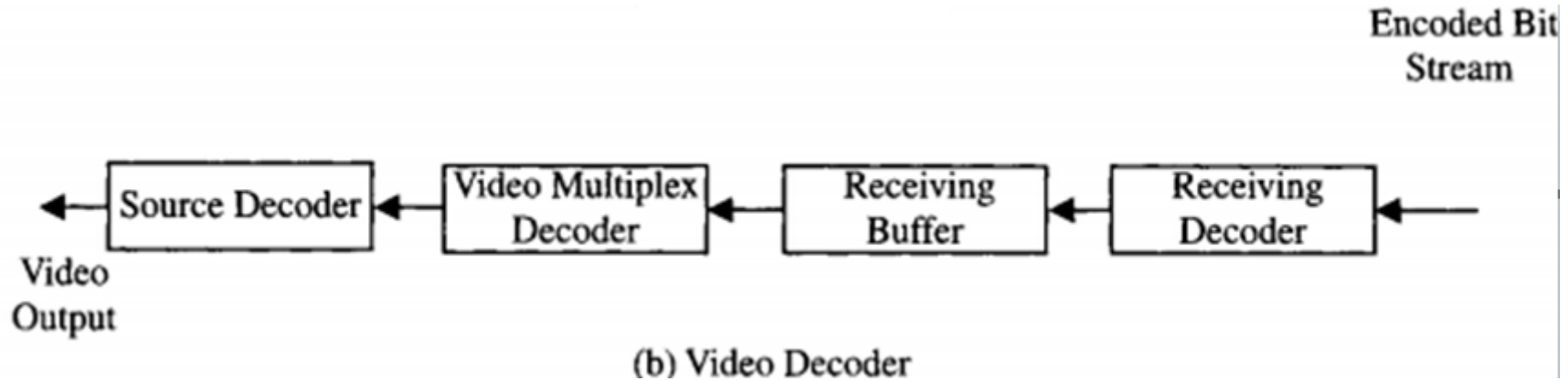
6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- The H.261 video encoder



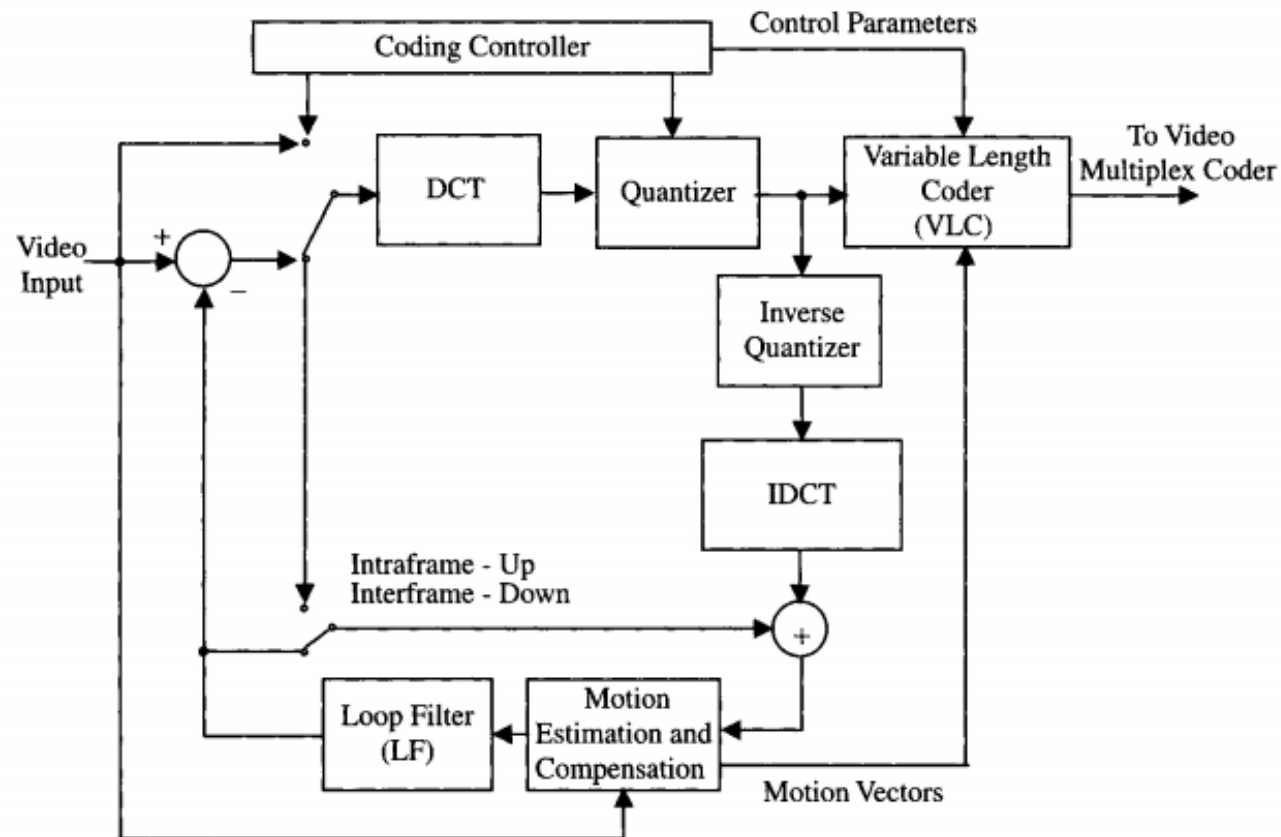
6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

- The H.261 video decoder



6.1 Basic Concepts and Techniques of Video Coding and the H.261 Standard

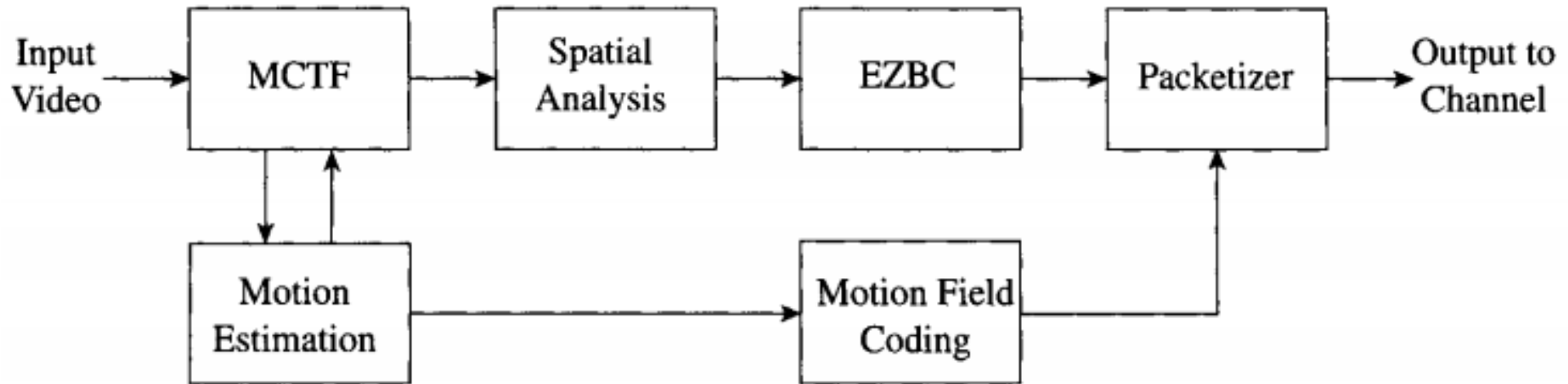
- H.261 Source Encode



(c) H.261 Source Encoder Implementation

6.2 Interframe Subband/Wavelet Scalable Video Coding

- MC-EZBC is one of a family of motion-compensated (MC) subband/wavelet coders that exploit temporal correlation but are fully embedded in quality/bit-rate, spatial resolution, and frame rate.



6.2 Interframe Subband/Wavelet Scalable Video Coding

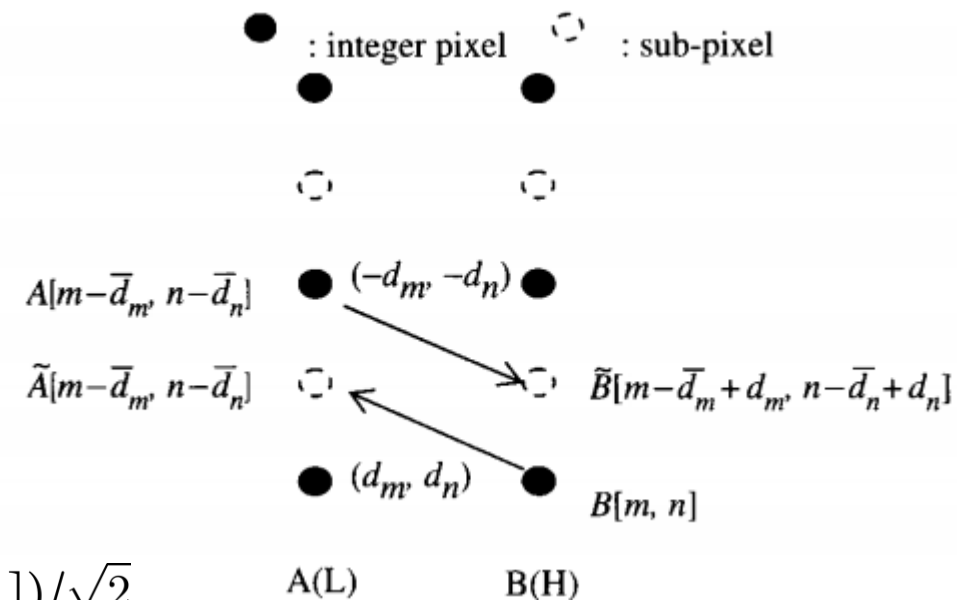
- Motion-compensated temporal filter (MCTF): essential role in motion compensated 3D SWT coding as affecting coding efficiency and temporal scalability
- Noninvertible approach

highpass coefficient

$$H[m, n] = (B[m, n] - \tilde{A}[m - d_m, n - d_n]) / \sqrt{2}$$

lowpass coefficient

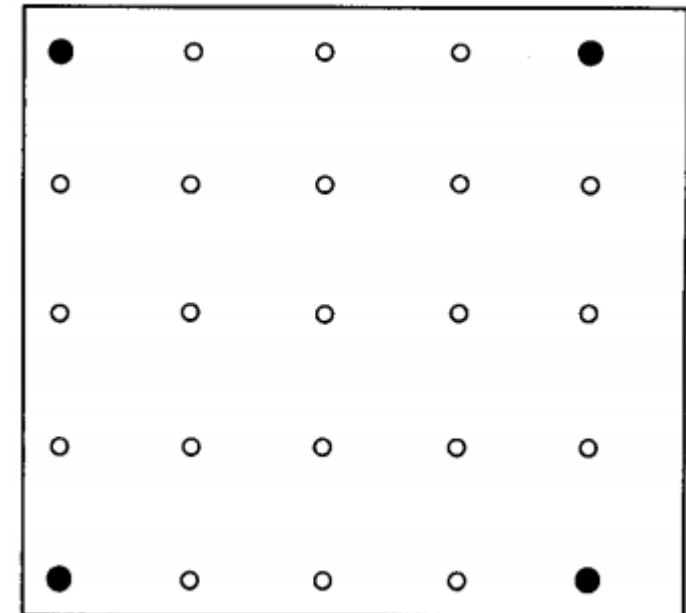
$$L[m - \bar{d}_m, n - \bar{d}_n] = (\tilde{B}[m - \bar{d}_m + d_m, n - \bar{d}_n + d_n] + A[m - \bar{d}_m, n - \bar{d}_n]) / \sqrt{2}$$



6.2 Interframe Subband/Wavelet Scalable Video Coding

- Motion-compensated temporal filter (MCTF): essential role in motion compensated 3D SWT coding as affecting coding efficiency and temporal scalability
- Lifting implementation: interpolated temporal highpass band was used to update A.

● : integer pixel ○ : interpolated subpixel

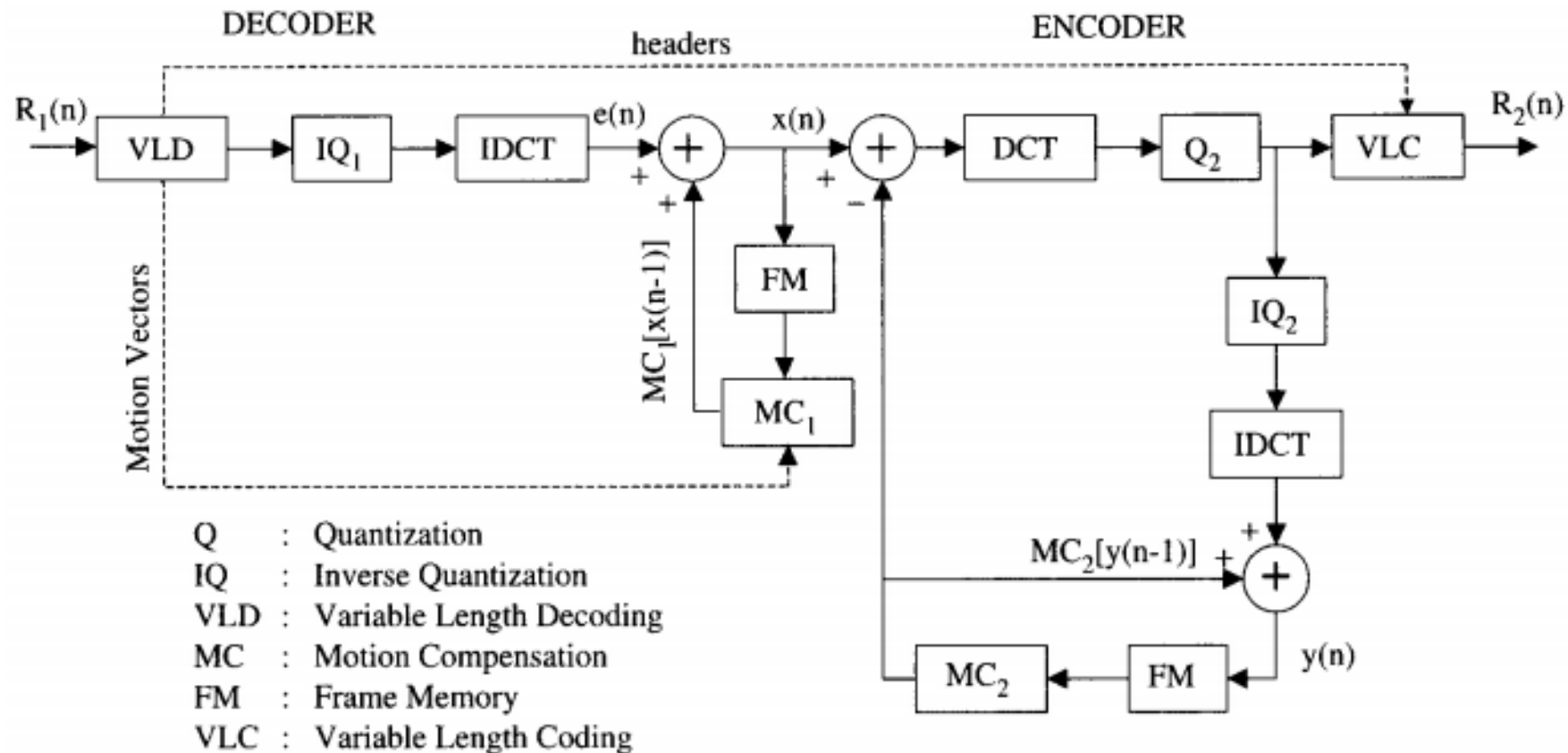


6.3 Digital Video Transcoding

- Video transcoding involves converting video from one compressed format to another to adapt to channel bandwidth or receiver requirements, facilitating video delivery across different networks
- It can be used for bit rate reduction, spatial and/or temporal resolution reduction, and video coding format conversion. Research on video transcoding initially focused on bit rate reduction to match available bandwidth
- With the rise of multiple video coding standards and new devices with limited capabilities, transcoding that changes format, resolution, and other parameters has become a key area of study.

6.3 Digital Video Transcoding – Video Transcoding for Bit Rate Reduction

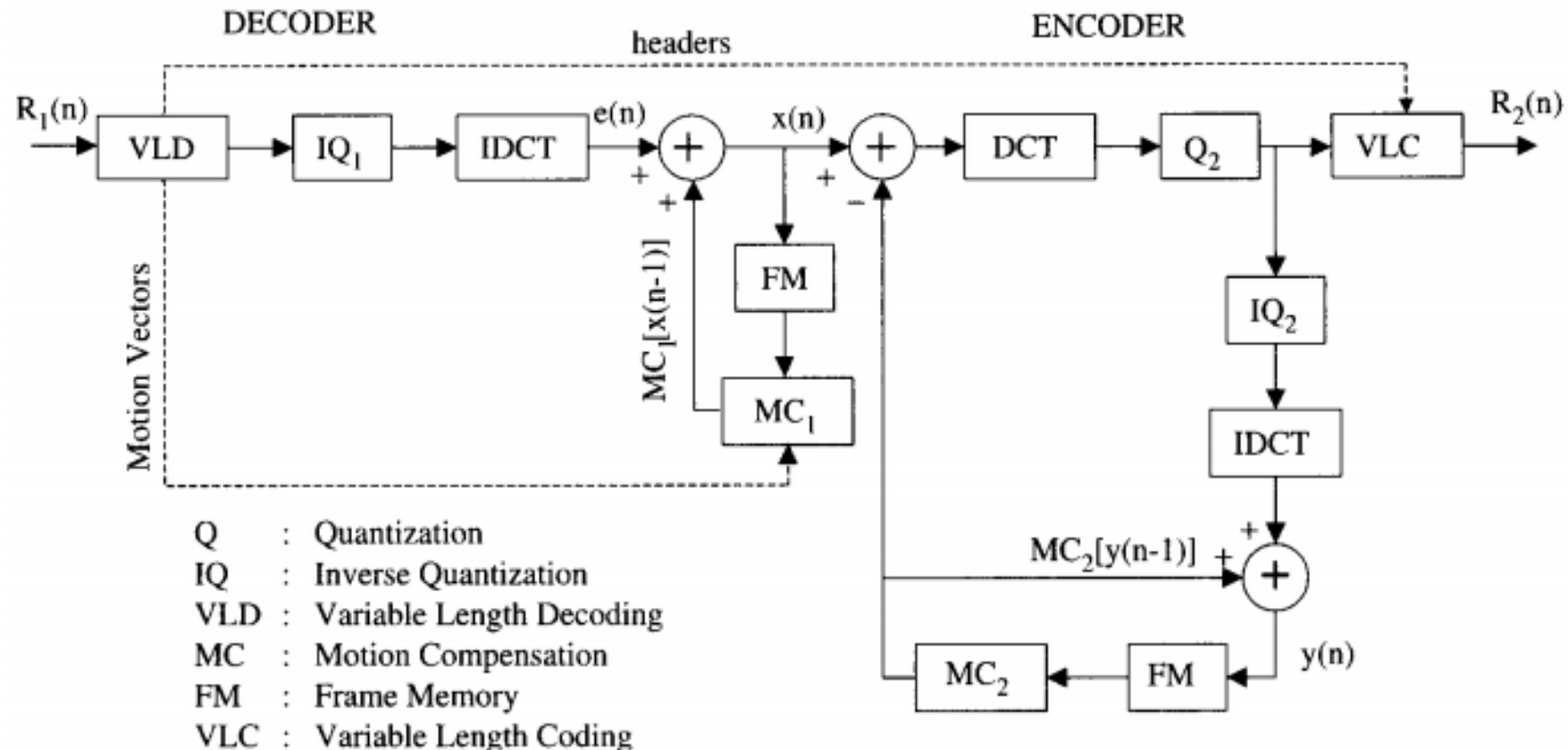
- Transcoding for intracoded frames $R_2(n) = Q_2[Q_1^{-1}(R_1(n))]$



6.3 Digital Video Transcoding – Video Transcoding for Bit Rate Reduction

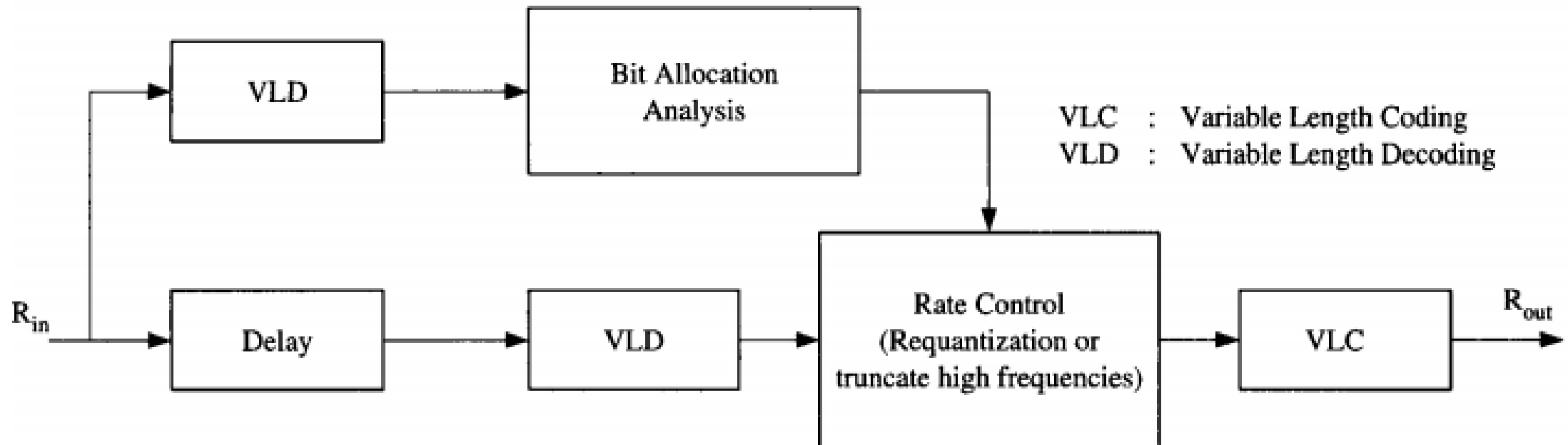
- Transcoding for intercoded frames

$$R_2(n) = Q_2[Q_1^{-1}[R_1(n) + DCT[MC_1(x(n-1)) - MC_2(y(n-1))]]]$$



6.3 Digital Video Transcoding – Video Transcoding for Bit Rate Reduction

- Fast video Transcoding Architectures

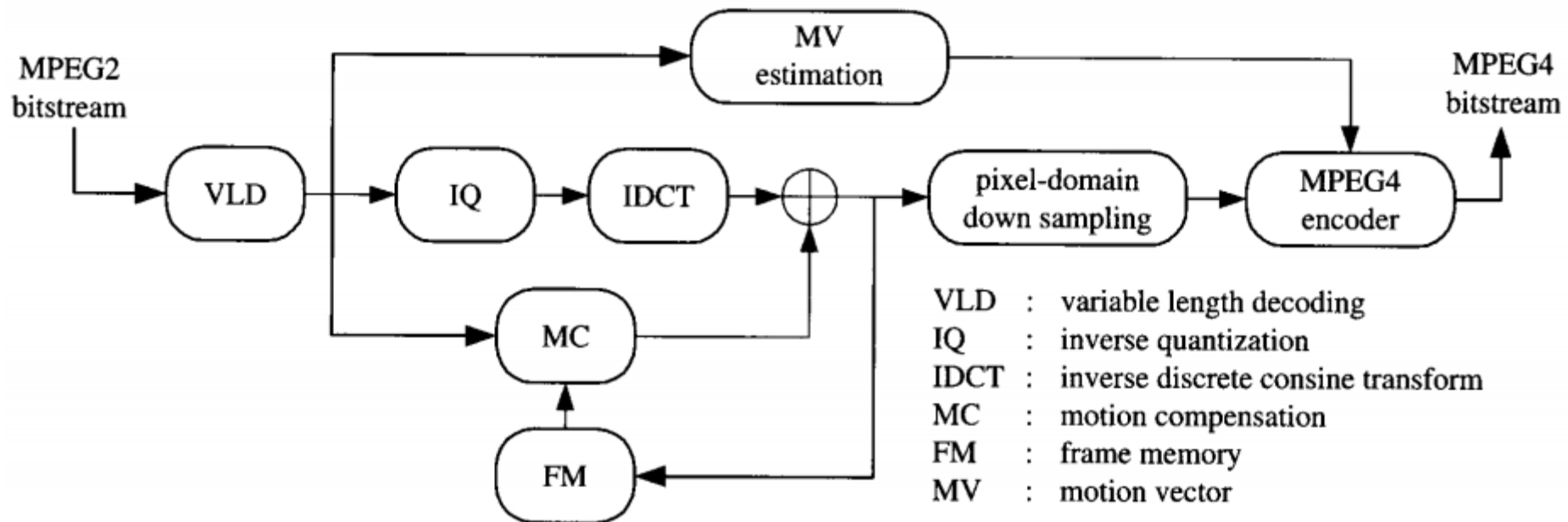


6.3 Digital Video Transcoding – Heterogeneous Video Transcoding

- Heterogeneous video transcoding enables universal multimedia access,
 - the coexistence of multiple video coding standards
 - the variety of devices with different characteristics used in networked multimedia applications
- The incoming video bitstream may need to be converted to a different coding format or have its spatial and/or temporal resolution reduced to meet the requirements of the communication channel or receiving devices
- Heterogeneous video transcoding requires separate decoder and encoder loops

6.3 Digital Video Transcoding – Heterogeneous Video Transcoding

- Pixel domain video transcoding architecture to transcode MPEG2 videos to MPEG4 bit stream with spatial and/or temporal resolution reduction

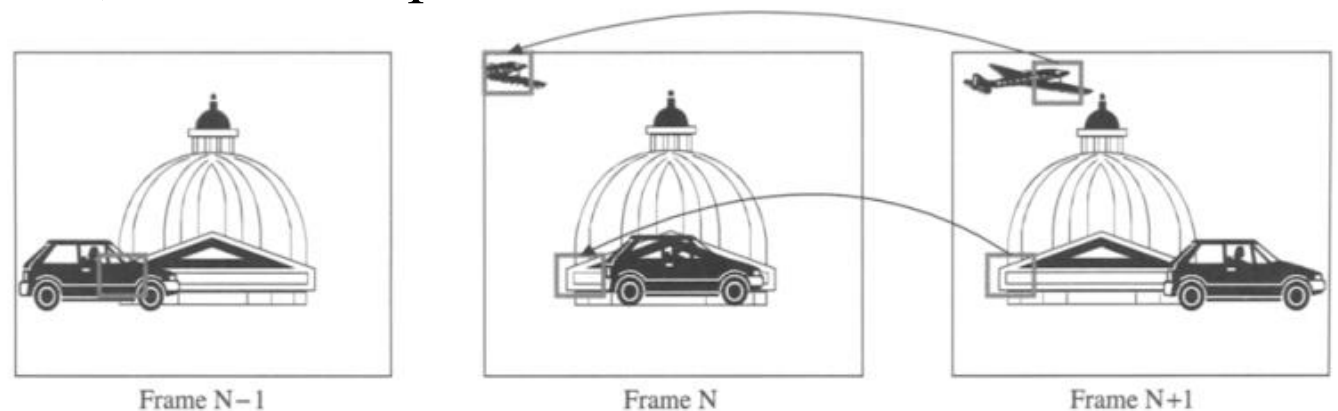


6.4 MPEG-1 and MPEG-2 Video Standards

- MPEG: Moving Picture Experts Group
- The development methodology of MPEG is divided into three phases
 - Requirements: set the focus of the effort and determine the rule for the competition phase Proposal Package Description and test methodology
 - Competition: obtain the state-of-the-art technology from academic and industrial research
 - Convergence: form one solution which implements a reference encoder and decoder

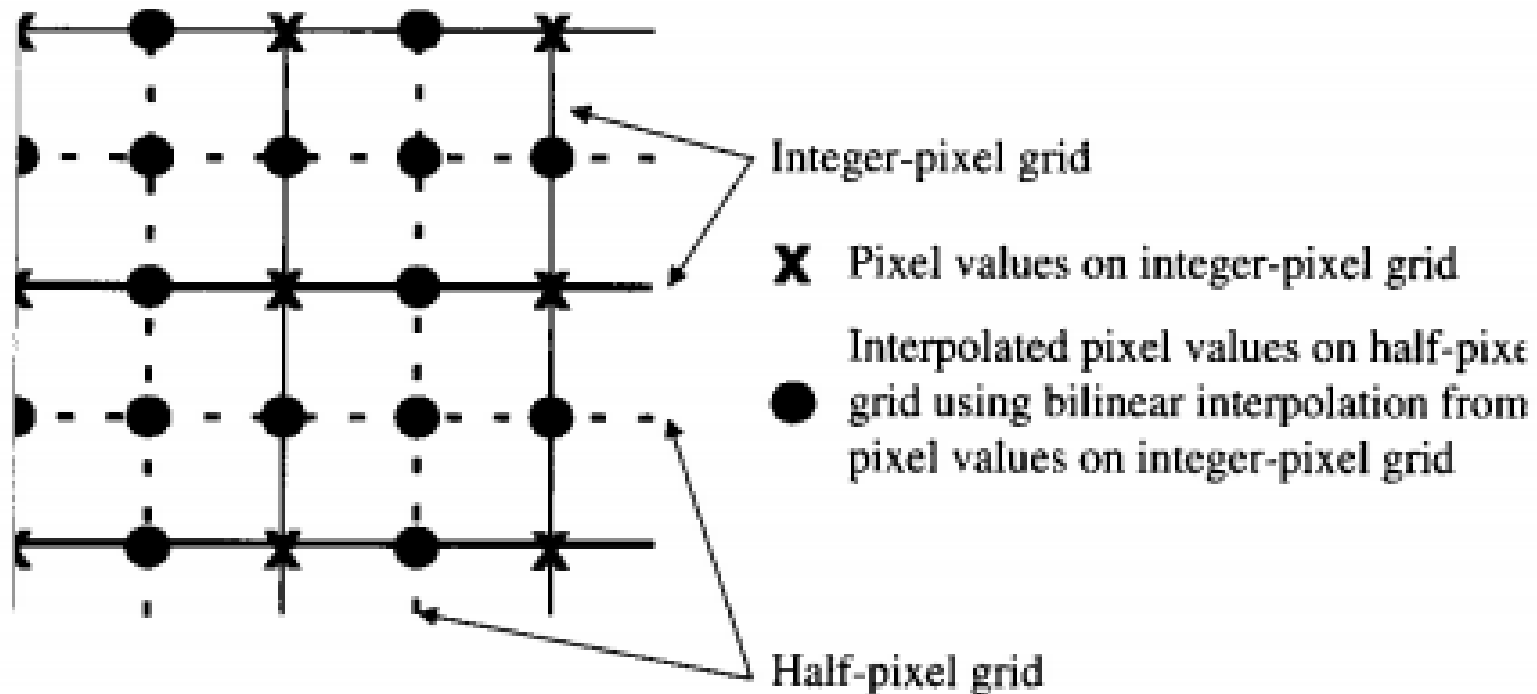
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Coding

- Bi-directional motion compensated prediction
 - MPEG-1 allows the future frame to be used as the reference frame for the motion compensated prediction (backward prediction) which can provide better prediction
 - The backward prediction can properly predict these uncovered areas since they are available in the future reference picture
 - If there are objects moving into the picture, these new objects cannot be predicted from the previous picture, but can be predicted from the future frame
 - Can help denoising



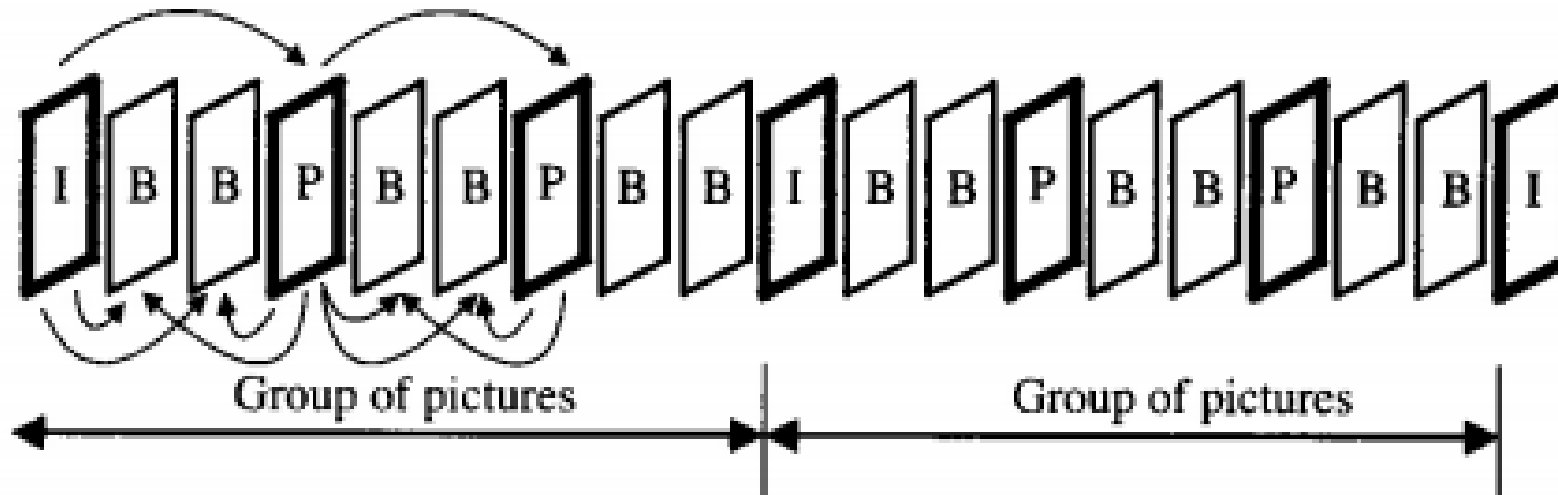
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Coding

- MPEG-1 allows half-pixel accuracy motion vectors → better than motion estimation with integer pixel accuracy



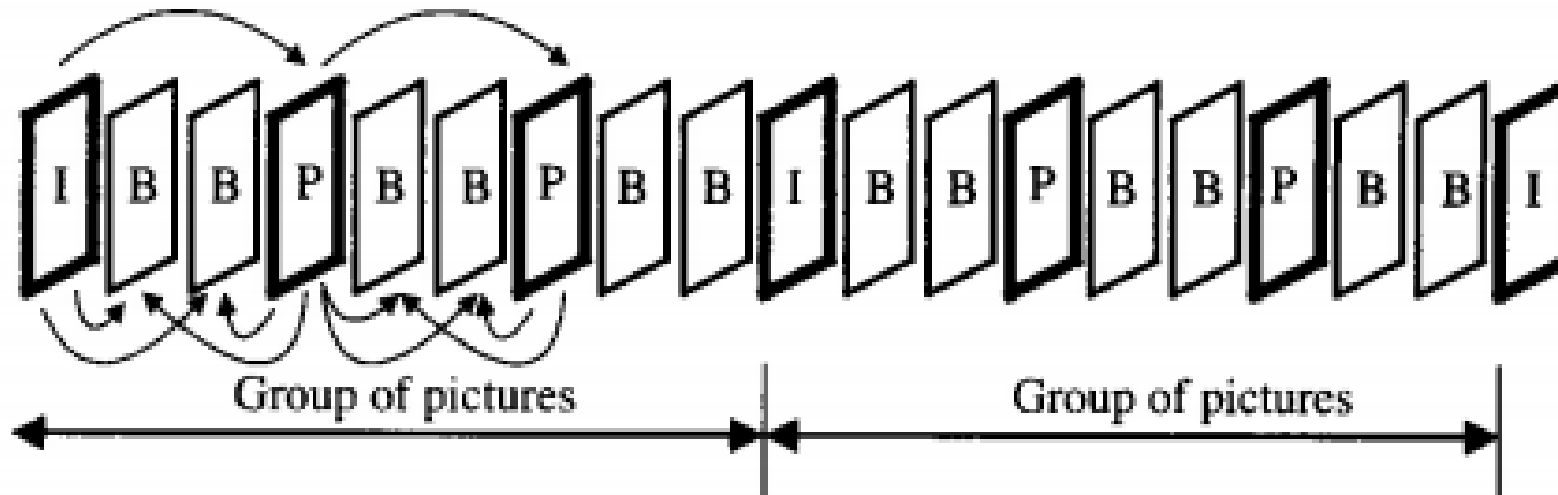
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Structure

- The typical MPEG-1 input format is the source input format (SIF).
- SIF is derived from ITU-R BT 601, formerly CCIR601, a worldwide standard for digital TV studio.
- In MPEG, each video sequence is divided into one or more groups of pictures (GOPs).



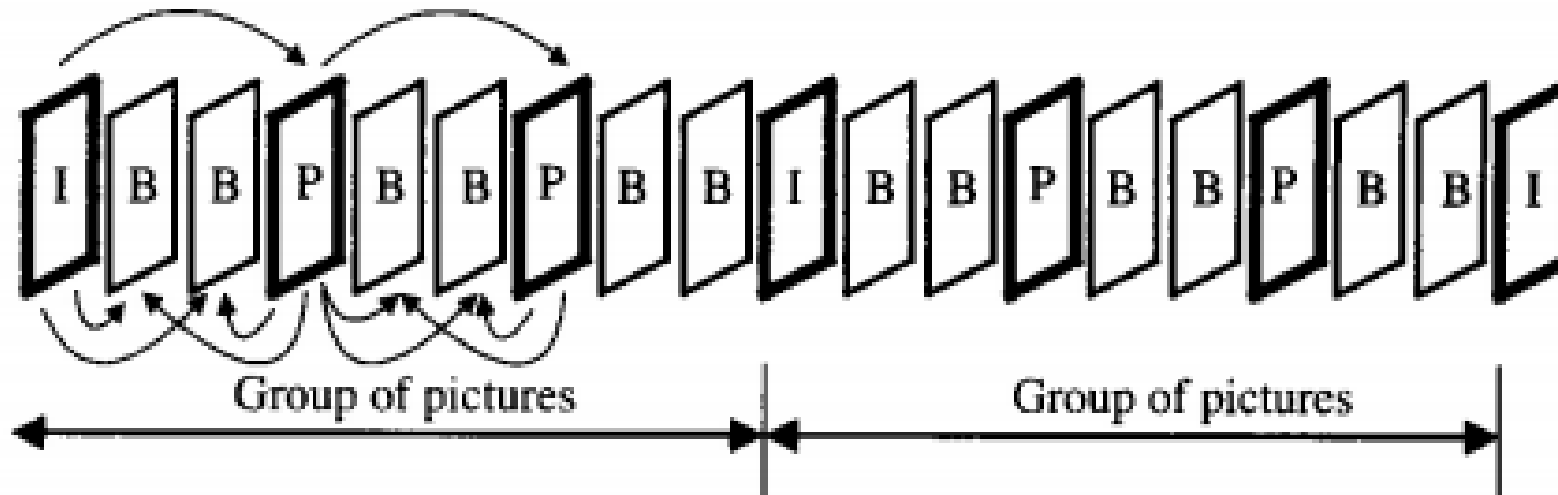
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Structure

- I-pictures (intra-coded pictures) are coded independently with no reference to other pictures. I-pictures provide random access points in the compressed video data, since the I-pictures can be decoded independently without referencing to other pictures.



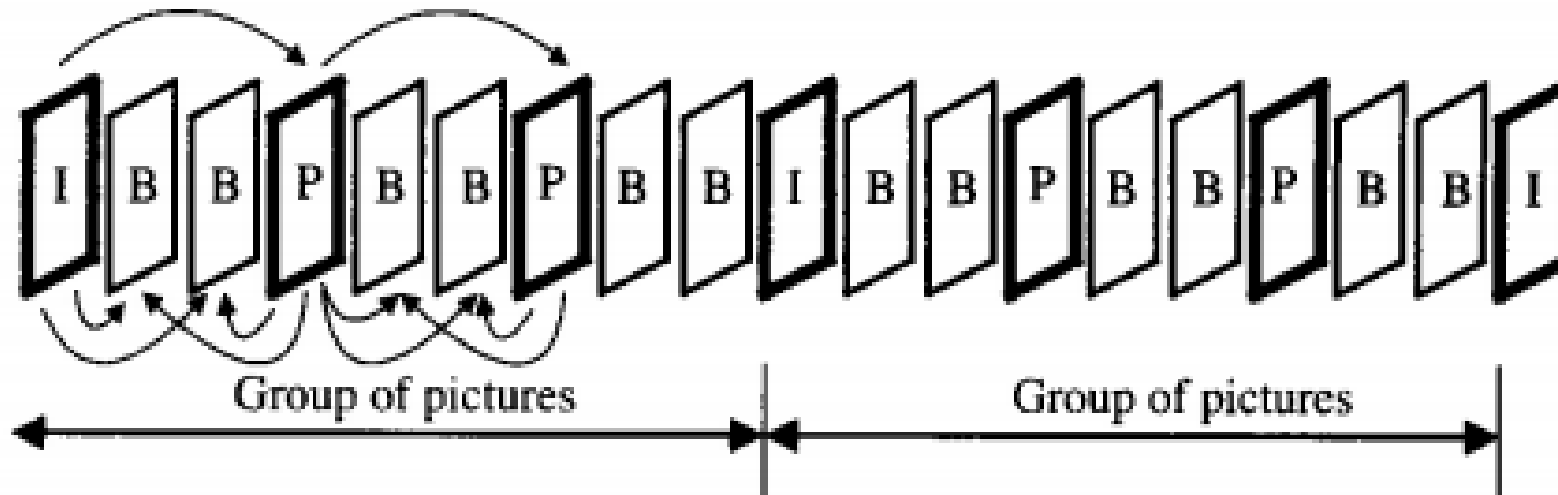
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Structure

- P-pictures (predictive-coded pictures) are coded using the forward motion-compensated prediction similar to that in H.261 from the preceding I- or P-picture. P-pictures provide more compression than the I-pictures by virtue of motion-compensated prediction.



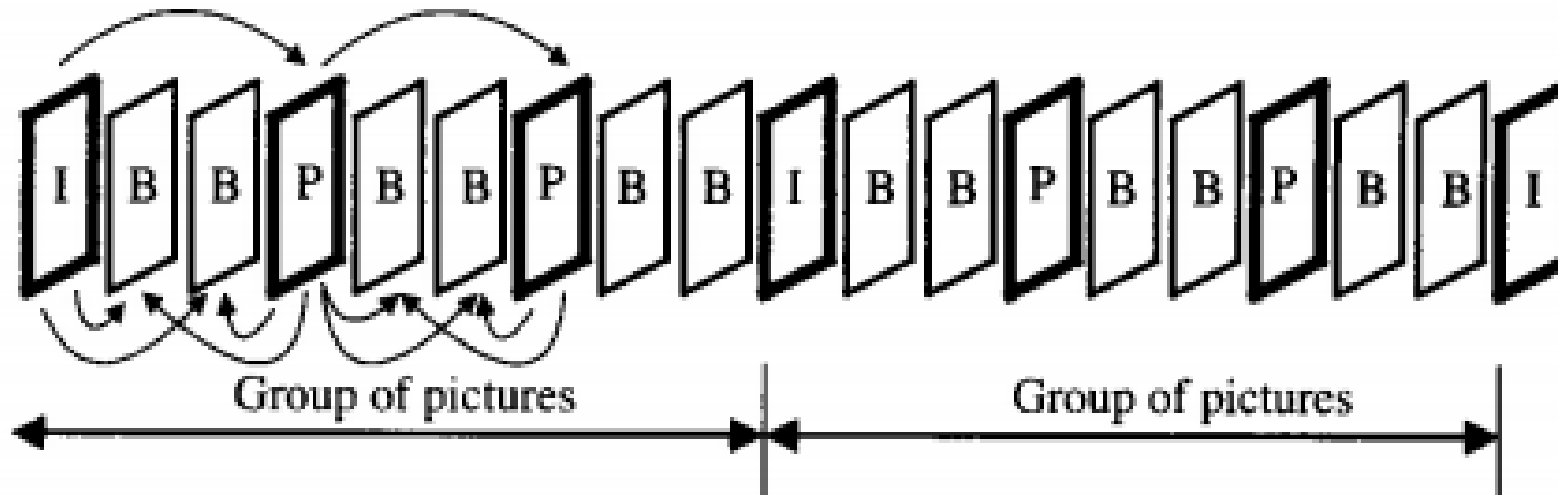
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Structure

- B-pictures (bi-directional predicted pictures) allow macro-blocks to be coded using bi-directional motion-compensated prediction from both the past and future reference I- or P-pictures.



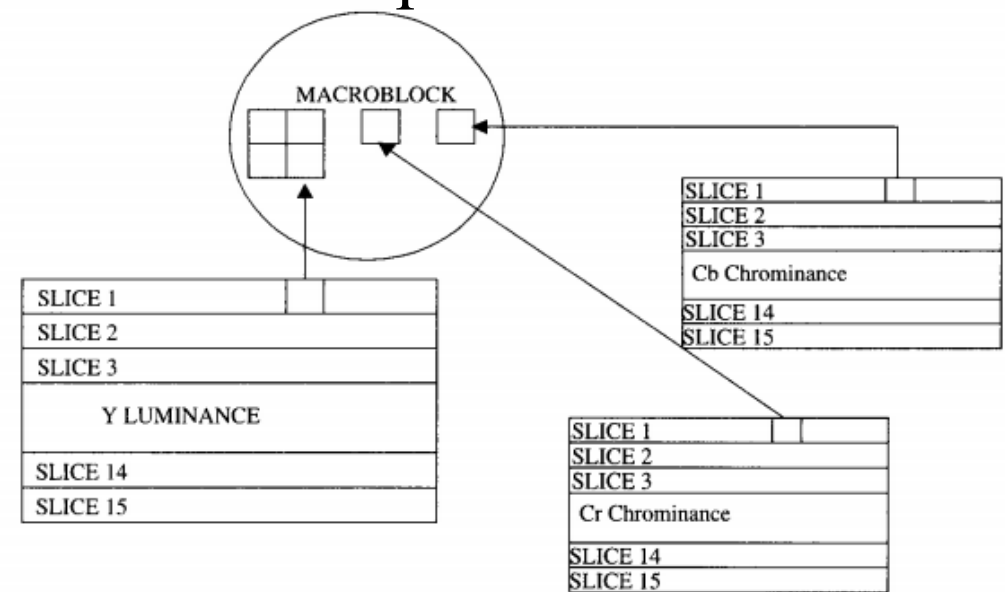
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Structure

- D-pictures (dc-pictures) are low-resolution pictures obtained by decoding only the dc coefficient of the discrete cosine transform (DCT) coefficients of each macroblock. They are not used in combination with I-, P-, or B-pictures. D-pictures are rarely used, but are defined to allow fast searches on sequential digital storage media.

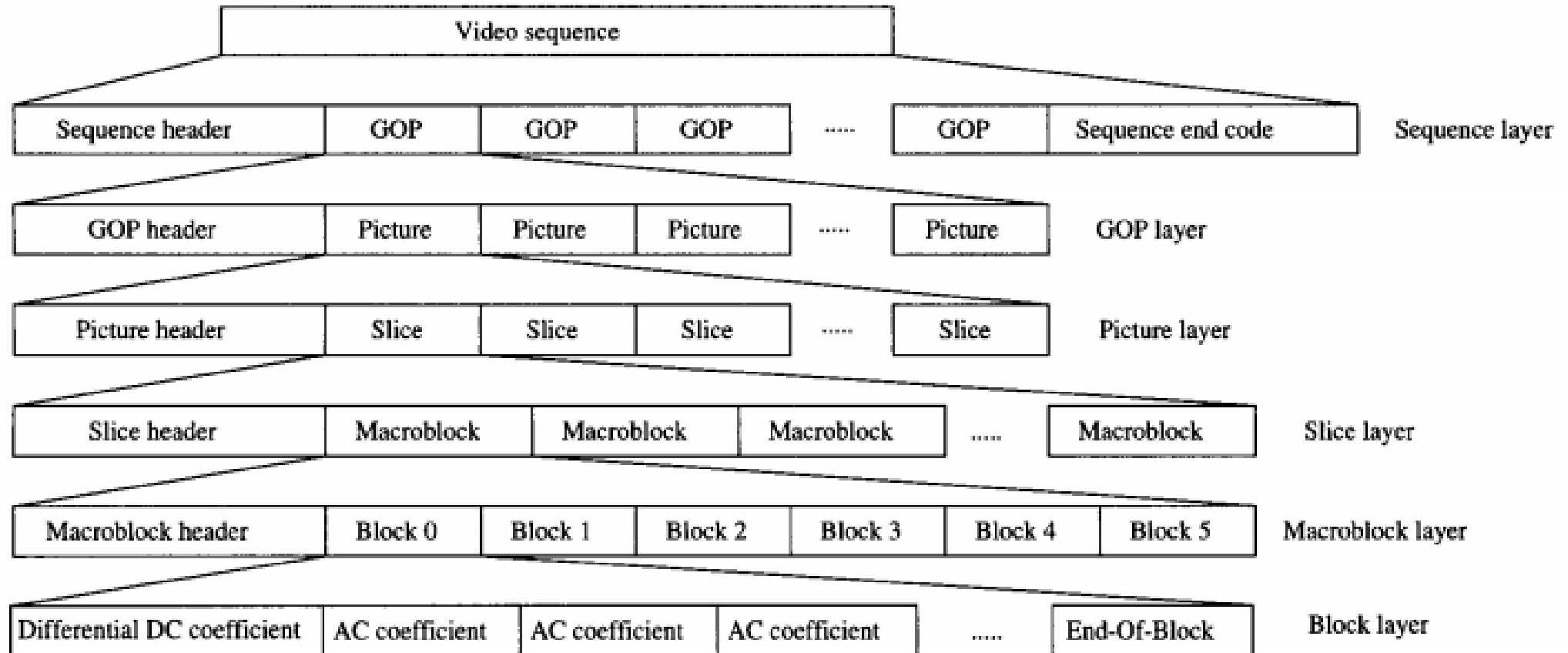


6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Structure

- An MPEG picture consists of slices. A slice consists of a contiguous sequence of macroblocks in a raster scan order from left to right and from top to bottom).
- A macroblock consists of a 16×16 block of luminance samples and two 8×8 blocks of corresponding chrominance samples



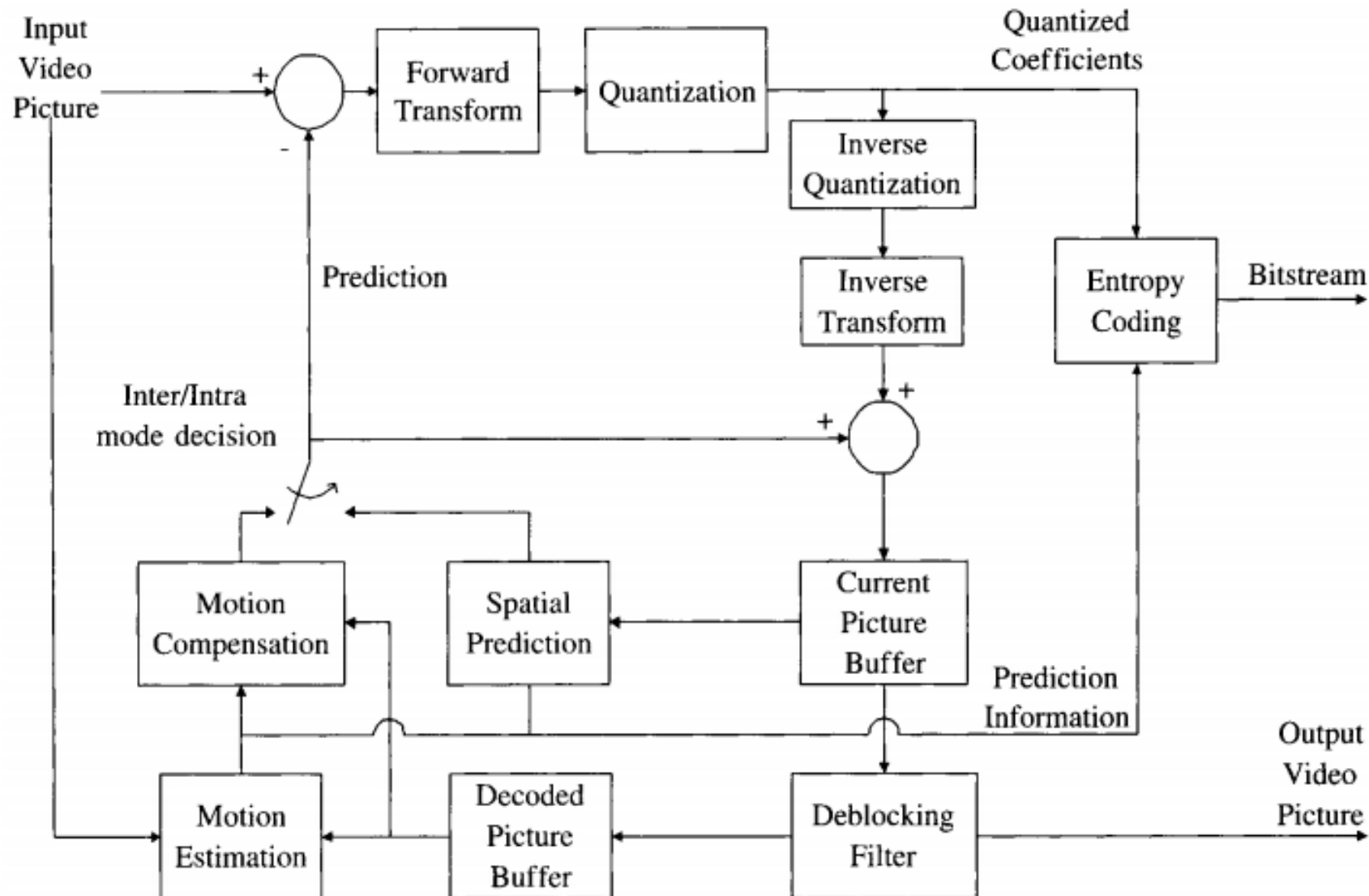
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-1 Video Bit Stream Structures



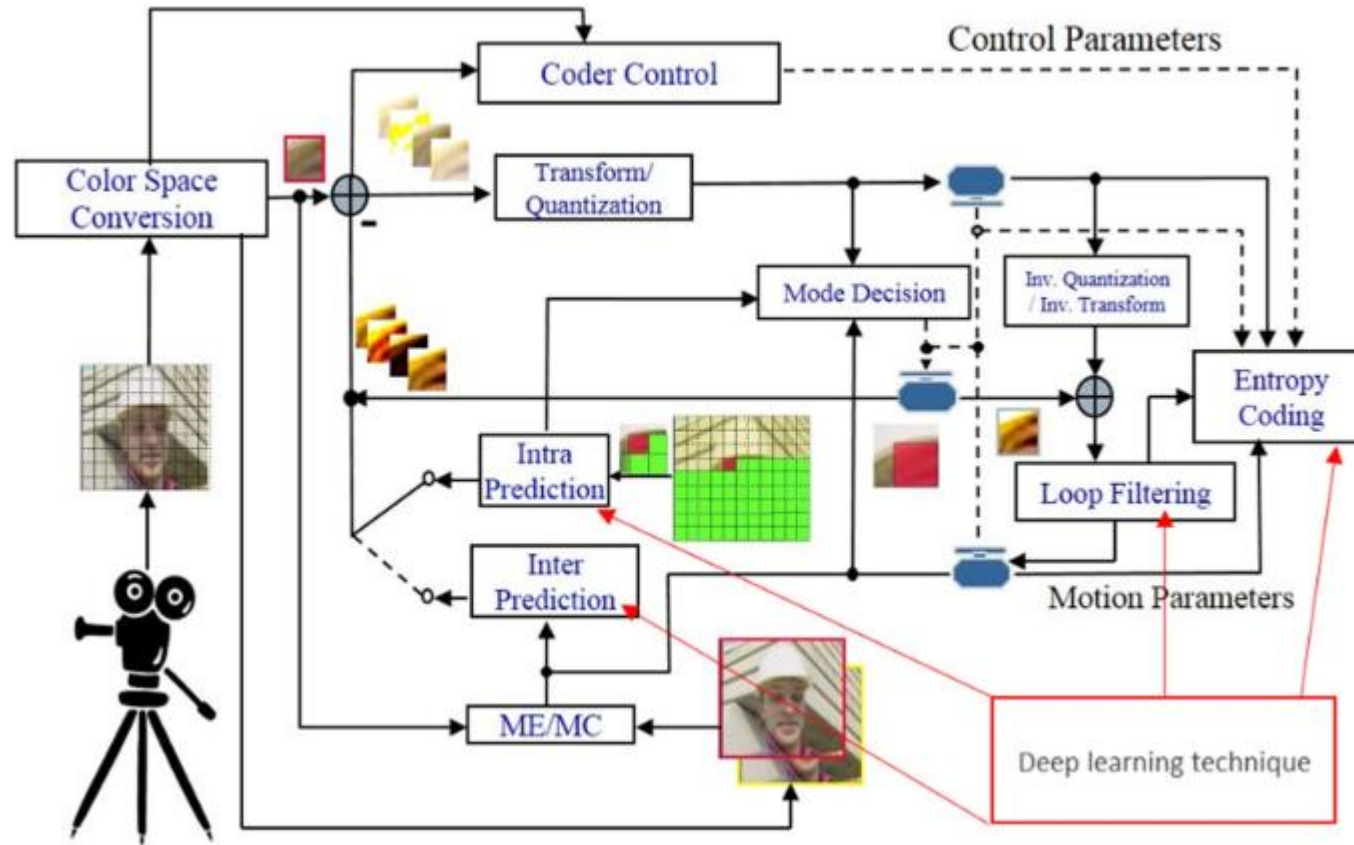
6.4 MPEG-1 and MPEG-2 Video Standards – MPEG-2 Video Coding

- Target applications
 - coding high-quality video at 4-15 Mb/s for video on demand (VOD), digital broadcast television, and Digital Storage Media such as DVD (digital versatile disc).
 - coding HDTV, cable/satellite digital TV, video services over various networks, 2-way communications, and other high-quality digital video applications
- Requirements
 - provide good NTSC quality video at a bit rate of about 4-6 Mbits/s and transparent NTSC quality video at a bit rate of about 8-10 Mbits/s.
 - provide the capability of random access and quick channel-switching by means of I-pictures in GOPs.

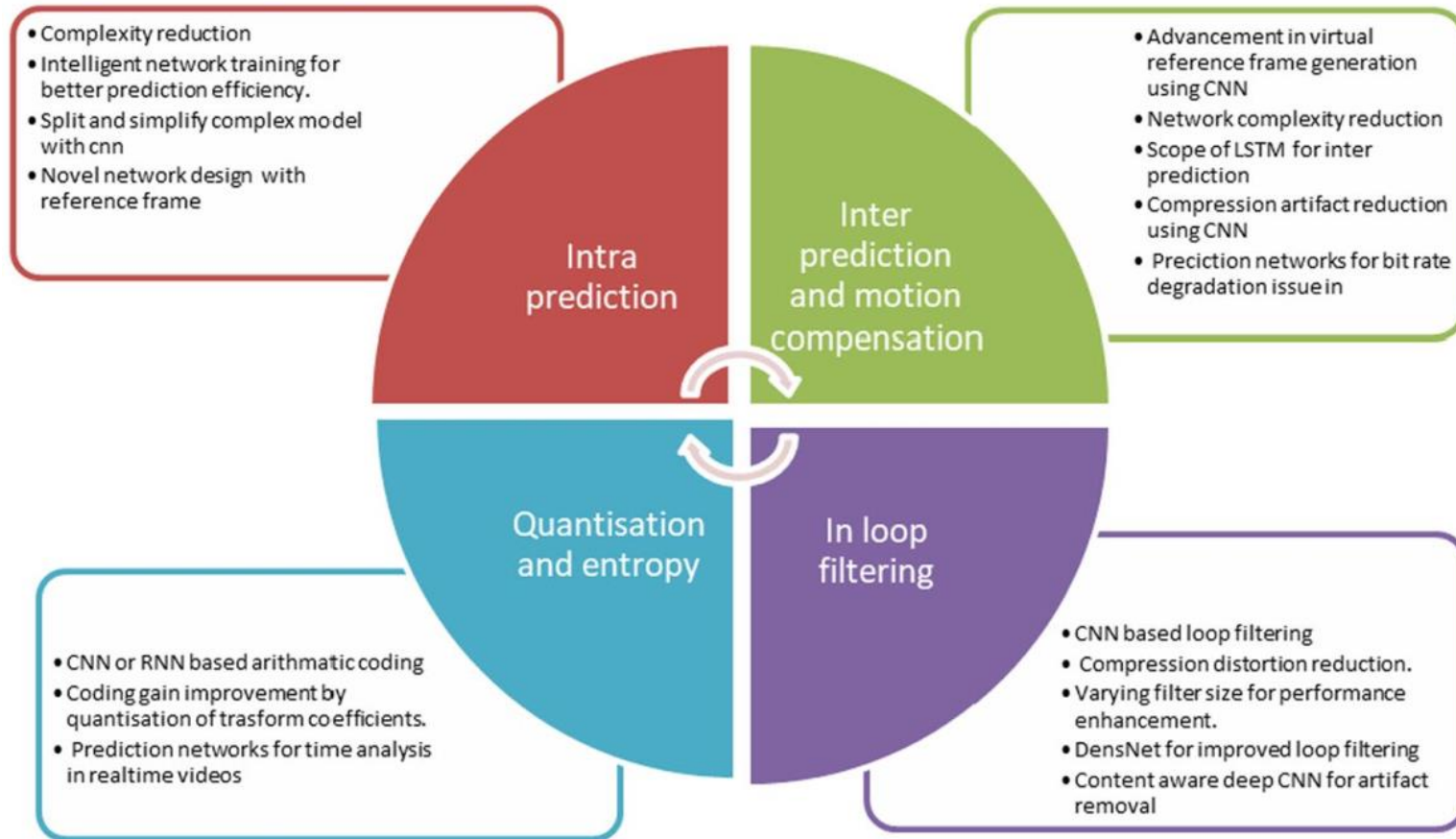
6.4 H.264/AVC Standard: Block diagram



6.5 Machine learning/deep learning based video compression



6.5 Machine learning/deep learning based video compression



Joy, H.K., Kounte, M.R., Chandrasekhar, A. *et al.* Deep Learning Based Video Compression Techniques with Future Research Issues. *Wireless Pers Commun* **131**, 2599–2625 (2023).

Summary

- Basic Concepts and Techniques of Video Coding and the H.261 Standard
- Interframe Subband/Wavelet Scalable Video Coding
- Digital Video Transcoding
- MPEG-1 and MPEG-2 Video Standards